

Main

Live door state. The spec name is Standard Operating Mode. The operator label is **Main**. This is the default screen after boot and after wake from a blanked panel.

Draft. Pixel layout: UI/index.html (320 × 170). LCD and web draw the same state tree. The web keypad sits to the right of the panel. Full chrome: overview figure 1 and figure 9 below.

Needs setup blocks automatic motion. Until origin, markers, and an armed obstruction table exist, hall and rocker triggers stay off. The header reads NEEDS SETUP. The hero is a red POSITION UNKNOWN banner. Run Calibration.

1. Live screen chrome

Every live Main state uses the same regions. A latched fault replaces the hero and the current meter with a red banner. GPIO numbers, duty, phase timers, bin index, and drift live in Diagnostics, not here.

Region	What it shows
Header left	MAIN when setup is complete, or NEEDS SETUP.
Header right	Link state. Green pip and STA plus the IP, or amber pip and AP plus the SoftAP name (default PrivateReserve). The join password is not in the header.
Hero	Current action in the largest type. A latched fault replaces this with a red banner.
Current	Live amps against the learned trip for the bin the door is in. The red tick is that bin's trip. The hard 2.50 A ceiling is the right edge. Hidden when a fault banner occupies the hero.
Travel	Share of the measured span in AS5600 counts. Ticks are the two near-end markers. Outer ends are open-seated and closed-seated. The grey line under the bar is percent and counts, or position unknown.
Tiles	Bottle key, open marker, close marker, shaft rev/s. The close-marker title clips to CLOSE MARK... on the 320-pixel panel.
Footer	TURN menu · PRESS menu. Either gesture opens the mode menu.

1.1 Needs setup — position unknown

Default after power loss until guided calibration finds origin. Automatic hall and rocker triggers stay off.

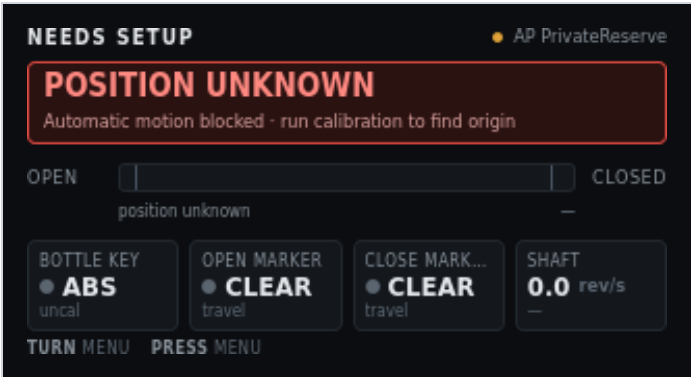


Figure 1. Main · needs setup. Position unknown. SoftAP serving.

On screen	Meaning
NEEDS SETUP	Header left. Origin, markers, or the calibration format is invalid. Automatic motion is blocked.
Amber pip + AP PrivateReserve	Header right. Station join failed or SSID is empty. Fallback SoftAP is serving. Default name is PrivateReserve. Default join password is reserved. The header does not show the password. Read it on Settings → Network or Diagnostics → Network. Then open http://192.168.4.1/ .
POSITION UNKNOWN	Red banner title. The AS5600 is single-turn. After reset the firmware does not know which turn the door is on.
Automatic motion blocked · run calibration to find origin	Banner detail. Hall and rocker triggers stay disabled until Calibration finds origin.
OPEN / CLOSED	Travel ends. Open is the left of the bar. Closed is the right.
Empty travel track, grey thumb	No fill and no marker ticks in this state. Position is not known, so the bar cannot show a share of span.
position unknown	Left scale under the bar. Replaces percent until origin exists.
– (right of the scale)	Count field. Empty because there is no origin to accumulate from.
BOTTLE KEY · grey pip · ABS · uncal	Hall reads absent. Key levels are not yet captured, so the grey line is uncal instead of an ADC count.
OPEN MARKER · grey pip · CLEAR · travel	Open near-end switch is not made. travel means the door is not at that end (or the end is unknown).
CLOSE MARK... · grey pip · CLEAR · travel	Close near-end switch is not made. Title is Close marker. The tile clips the word.
SHAFT · 0.0 rev/s · –	Motor at rest. Direction dash means no rotation this cycle.
TURN menu · PRESS menu	Either gesture opens the mode menu. Position Unknown is not an acknowledge-away fault.

1.2 Opening

Setup is complete. The bottle key went absent. The door is in cruise toward open.

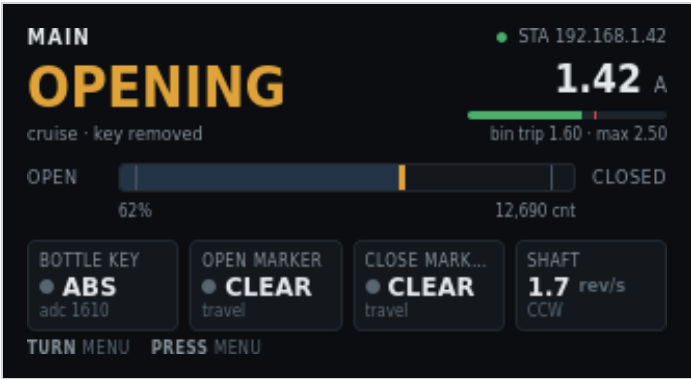


Figure 2. Main · opening. Cruise, key removed.

On screen	Meaning
MAIN	Header left. Setup is complete. Automatic motion is allowed.
Green pip + STA 192.168.1.42	Header right. ESP32 joined site Wi-Fi as a station. HTTP is on this address.
OPENING (amber)	Hero. Current action. Amber means the motor is moving.
cruise · N in left	Motion phase and remaining throw to the destination marker. The bottle key is absent.
1.42 A	Live motor current from the ACS712 after scale.
Green fill on the current bar	Live amps as a share of the 2.50 A hard ceiling. Green because 1.42 A is under this bin's trip.
Red tick on the current bar	This bin's learned trip (1.60 A). Not the hard ceiling.
bin trip 1.60 · max 2.50	Numeric trip for the bin the door is in, then the compile-time ceiling at the right edge of the bar.
OPEN / CLOSED	Travel ends.
Blue fill to 62%, yellow thumb, two marker ticks	Door is 62% of the measured span toward closed. Marker ticks sit near each end. Neither tick is hot.
62%	Share of measured span. Percent is the headline. Counts sit on the right.
12,690 cnt	AS5600 counts from the open origin.
BOTTLE KEY · grey pip · ABS · adc 1610	Key absent. Raw ADC is the diagnostic value for this tile.
OPEN MARKER · grey pip · CLEAR · travel	Open switch not made. Door is in mid travel.
CLOSE MARK... · grey pip · CLEAR · travel	Close switch not made.
SHAFT · 1.7 rev/s · CCW	Shaft speed and direction. On this build CCW is toward open.
TURN menu · PRESS menu	Open the mode menu. There is no Stop on Main.

1.3 Idle — closed and seated

Move finished. Seal is loaded. Motor is at rest.

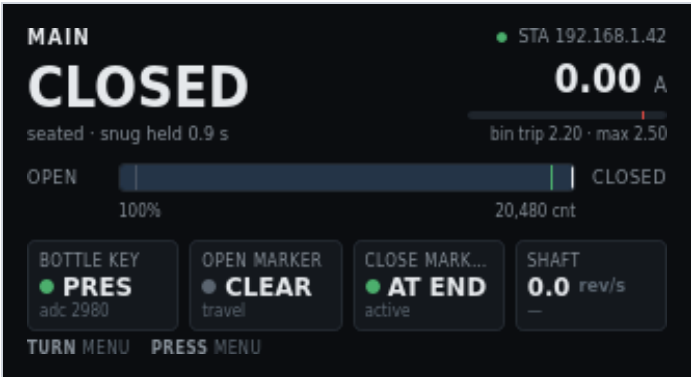


Figure 3. Main · idle closed. Seated, snug held.

On screen	Meaning
MAIN	Header left. Setup complete.
Green pip + STA 192.168.1.42	Station Wi-Fi, HTTP on this IP.
CLOSED	Hero. Door is at the closed-seated end. White, not amber, because the motor is idle.
seated · snug held 0.9 s	How the stop was held. Snug is current-rise seating past the close marker. 0.9 s is hold time this cycle.
0.00 A	Motor current at rest.
Empty current bar, red tick near the right	Live fill is zero. The tick is this bin's trip (2.20 A).
bin trip 2.20 · max 2.50	Closed-end bin trip, then the 2.50 A ceiling.
OPEN / CLOSED	Travel ends.
Fill to the closed end, close-marker tick hot	Door is at 100% of span. The close-marker tick is active because the switch is made.
100% · 20,480 cnt	At closed seated. Counts include travel past the close-marker edge.
BOTTLE KEY · green pip · PRES · adc 2980	Bottle in the recess. Green pip means present on this polarity.
OPEN MARKER · grey pip · CLEAR · travel	Open switch not made.
CLOSE MARK... · green pip · AT END · active	Close switch is made. AT END is the cooked label. active is the raw switch state.
SHAFT · 0.0 rev/s · —	At rest. No rotation this cycle.
TURN menu · PRESS menu	Open the mode menu.

1.4 Closing — creep and snug

Past the close marker. The profile has left cruise and is seating the seal.

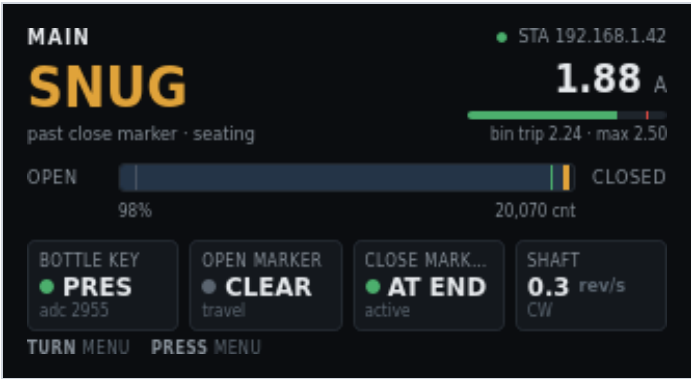


Figure 4. Main · snug. Past close marker, seating.

On screen	Meaning
MAIN	Header left.
Green pip + STA 192.168.1.42	Station Wi-Fi.
SNUG (amber)	Hero. Creep-and-seat phase after the close marker. Motor is still moving.
past close marker · seating	Why snug is showing. Operational snug is a current-rise stop, not a home.
1.88 A	Live current. Seal loading raises amps above cruise.
Green fill, red tick near the right	1.88 A is still under this bin’s 2.24 A trip.
bin trip 2.24 · max 2.50	Near-closed bin trip, then the hard ceiling.
OPEN / CLOSED	Travel ends.
Fill at 98%, close-marker tick hot	Almost seated. Close switch is made.
98% · 20,070 cnt	Share of span and counts from origin.
BOTTLE KEY · green pip · PRES · adc 2955	Bottle present. Close was commanded with the key in.
OPEN MARKER · grey pip · CLEAR · travel	Open switch not made.
CLOSE MARK... · green pip · AT END · active	Close switch made.
SHAFT · 0.3 rev/s · CW	Slow creep toward closed. CW is toward closed on this build.
TURN menu · PRESS menu	Open the mode menu. The move ends at seated, a reverse trigger, a fault, or obstruction handling.

1.5 Obstruction — reversing

Close current crossed the bin trip. Firmware reversed toward open. Obstruction handling outranks ordinary commands.

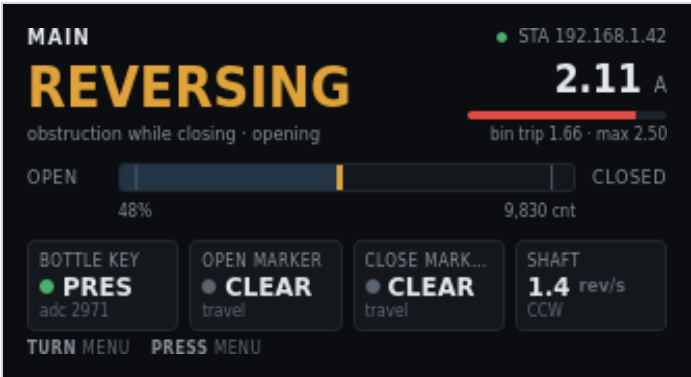


Figure 5. Main · reversing. Obstruction while closing.

On screen	Meaning
MAIN	Header left.
Green pip + STA 192.168.1.42	Station Wi-Fi.
REVERSING (amber)	Hero. Close obstruction. The door is opening to clear the path.
obstruction while closing · opening	Cause and the commanded direction now.
2.11 A	Live current. Above this bin's trip, so the fill is red.
Red fill past the red tick	Live amps exceeded the learned trip (1.66 A). That is why reverse started.
bin trip 1.66 · max 2.50	Mid-travel bin trip, then the hard ceiling. Reverse used the learned trip, not the 2.50 A ceiling.
OPEN / CLOSED	Travel ends.
Fill at 48%, neither marker hot	Mid travel. Both switches clear.
48% · 9,830 cnt	Position when reverse is in progress.
BOTTLE KEY · green pip · PRES · adc 2971	Bottle still present. Close had been the automatic command.
OPEN MARKER · grey pip · CLEAR · travel	Not yet at the open end.
CLOSE MARK... · grey pip · CLEAR · travel	Not at the close end.
SHAFT · 1.4 rev/s · CCW	Opening. CCW toward open on this build.
TURN menu · PRESS menu	Open the mode menu. Reverse still owns the motor until it finishes or faults.

1.6 Fault — second close obstruction

A second close obstruction in the retry window latches a fault. The current meter leaves the panel. PRESS acknowledges when firmware allows it.

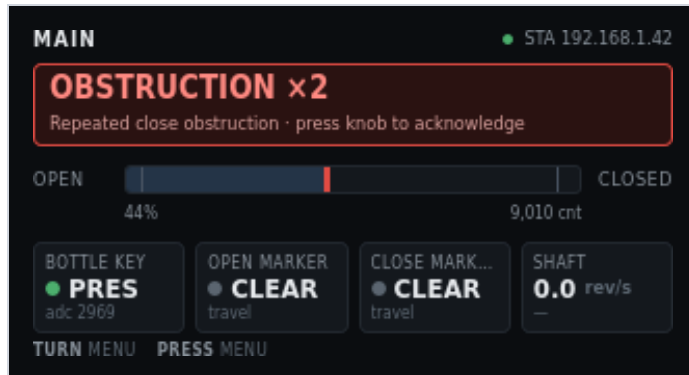


Figure 6. Main · latched obstruction fault. Press to acknowledge.

On screen	Meaning
MAIN	Header left. Setup is still complete. This is a motion fault, not Needs Setup.
Green pip + STA 192.168.1.42	Station Wi-Fi.
OBSTRUCTION x2	Red banner title. Second close obstruction. The door stopped instead of reversing again.
Repeated close obstruction · press knob to acknowledge	Banner detail. PRESS clears this banner when the firmware allows it. TURN still opens the mode menu.
OPEN / CLOSED	Travel ends. The current meter is hidden while the fault banner occupies the hero.
Red thumb at 44%	Position where the second trip stopped the door. Red track because this is a fault state.
44% · 9,010 cnt	Stopped position.
BOTTLE KEY · green pip · PRES · adc 2969	Bottle still present.
OPEN MARKER · grey pip · CLEAR · travel	Not at the open end.
CLOSE MARK... · grey pip · CLEAR · travel	Not at the close end.
SHAFT · 0.0 rev/s · —	Motor stopped after the fault.
TURN menu · PRESS menu	On this fault, PRESS acknowledges. TURN opens the mode menu. Position Unknown is the fault that cannot be acknowledged away.

2. Mode menu

TURN or PRESS on the live Main screen opens this list. Five items. TURN scrolls. PRESS opens. HOLD returns to the live Main screen.

2.1 Five destinations

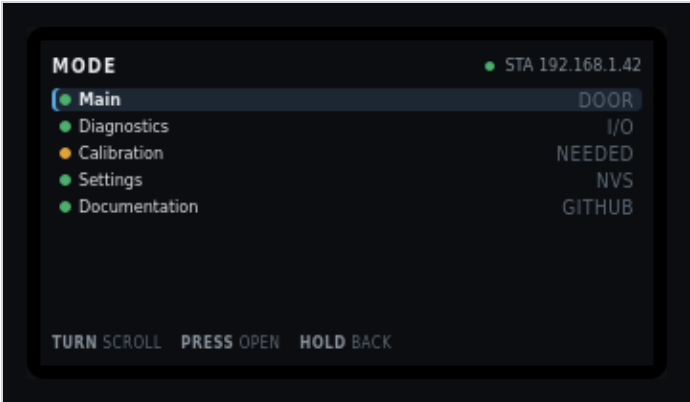


Figure 7. Mode menu. Five items. Calibration pip is amber because setup is still needed. Documentation is the fifth row.

On screen	Meaning
MODE	Header left. Same screen family as Main. Not a fourth operating mode.
Green pip + STA 192.168.1.42	Header right. Same link state as Main.
Blue highlight + left bracket on Main	Selected row. PRESS opens the live Main screen.
Green pip + Main + DOOR	Live door. Value column is the subject of that page.
Green pip + Diagnostics + I/O	Opens the diagnostic menu.
Amber pip + Calibration + NEEDED	Opens the stage list. Amber pip and NEEDED mean origin or later stages are missing. After a full save this row shows green and a status such as armed.
Green pip + Settings + NVS	Opens settings groups. Value is NVS when stored, or unsaved when staged edits exist. If the motor is running, Settings Locked instead.
Green pip + Documentation + GitHub	Opens the Documentation page. QR code for the public project URL.
TURN scroll	Move the highlight.
PRESS open	Enter the highlighted item.
HOLD back	Return to the live Main screen without changing mode.

Web nav tabs send the same jumps: Main, Calibration, Diagnostics, Settings, Docs. They turn the LCD on if it was blank. Settings is a heavier first paint (group list plus a possible NVS write when leaving jog); the firmware feeds the watchdog on that path so HTTP does not die after the list shows.

2.2 Documentation

Not an operating mode. A read-only page from the mode menu. Header is DOC. The panel shows a QR code for the public project page and the same path as text. This page does not command the motor. PRESS does nothing. HOLD returns to the mode menu.

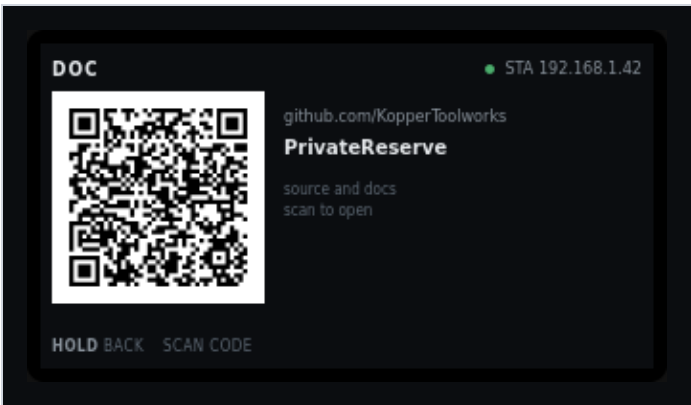


Figure 8. DOC. QR and path for the public GitHub page.

On screen	Meaning
DOC	Header left. Documentation page.
Green pip + STA 192.168.1.42	Header right. Same link state as Main. Amber pip and AP when the fallback SoftAP is serving.
QR code (left)	Encodes https://github.com/KopperToolworks/PrivateReserve. Scan with a phone camera. The enclosure sticker uses the same URL.
github.com/KopperToolworks	Host, as text, if the QR is hard to scan.
PrivateReserve	Repository name.
source and docs	What the page holds. Source, operator docs, and enclosure notes.
scan to open	How to use the QR. On the web copy of this screen, a blue link labelled Open GitHub page opens the same URL. The full web chrome (tabs and keypad) is overview figure 1 and figure 9 below.
HOLD back	Return to the mode menu.
SCAN code	Footer hint. Not a knob gesture. PRESS on this page does nothing.

Pixel source: Main mockup (Documentation state). The enclosure sticker uses the same URL; see DIN/enclosure.md. The repo must stay public or the QR is useless.

3. Knob on Main

Gesture	Action
TURN or PRESS (live screen)	Open the mode menu.
PRESS on Documentation	No-op. The QR is the action.
HOLD on Documentation	Return to the mode menu.
PRESS on a latched fault that can be acknowledged	Clear the banner when the firmware allows it. Position Unknown is not an acknowledge-away fault. It needs calibration.
Hall / rocker	Open or close when setup is ready. Edges, not hold-to-run. Open wins if both rocker throws hit in the same cycle.

There is no Stop on this screen. A move ends at destination, snug, a reverse trigger, a fault, or obstruction handling.

4. Web page

The LCD shows only the 320 × 170 panel. The phone page adds tabs on top and a keypad on the right. Use landscape. The shell scales to the viewport. There is no application login.

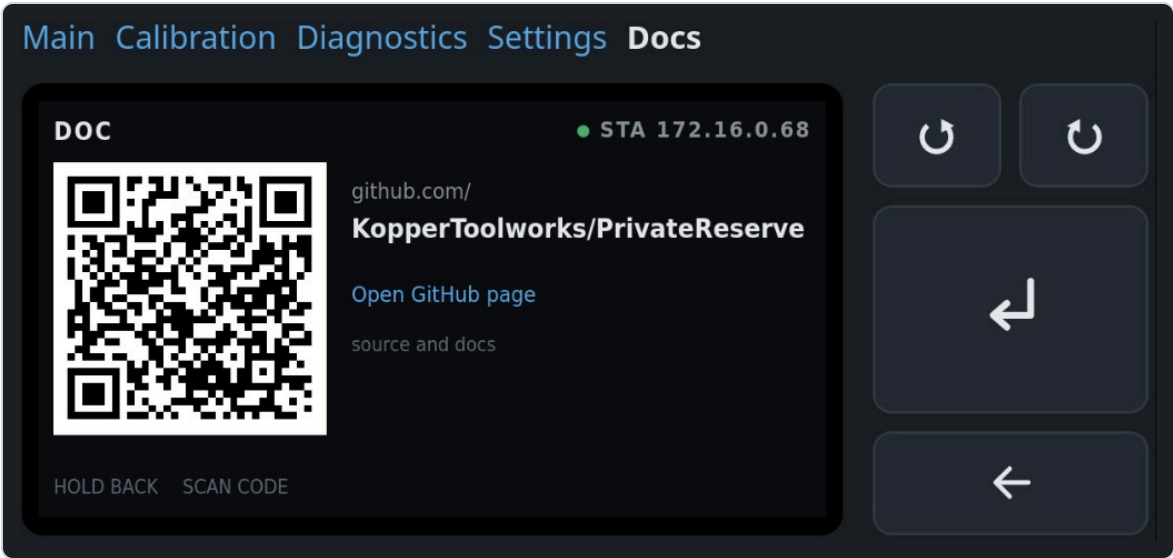


Figure 9. Full web UI on Documentation. Same chrome on every page. Overview figure 1 names every chrome label.

Chrome	Command
Tabs: Main · Calibration · Diagnostics · Settings · Docs	Jump to that page. Active tab is white. They turn the LCD on if it was blank.
Left circular arrow	TURN – (matches CCW on the physical knob)
Right circular arrow	TURN + (matches CW on the physical knob)
Large Return arrow	PRESS
Wide left arrow	HOLD back

On Documentation the web panel also shows Open [GitHub](#) page. That link is not on the LCD. PRESS on this page does nothing. HOLD returns to the mode menu.

5. Related

- [Software overview](#)
- [Settings](#) — clear station Wi-Fi and reboot
- [Board A](#) — terminals named on the tiles

This page is a chapter of the Software User Manual.